MIL-R-36982
26 June 1973

MILITARY SPECIFICATION

RETRACTOR, ABDOMINAL, BALFOUR, SELF-RETAINING, 6 BLADES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-retaining, abdominal retractor of the Balfour type.

1.2 Classification. The retractors covered by this specification shall be of the following styles as specified:

Style 1
Style 2

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-up.

FSC 6515

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Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)
No. 1286 Retractor, Abdominal, Balfour, Self-Retaining
No. 20586 Retractor, Balfour

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General Requirements. The abdominal retractor shall be in accordance with the applicable requirements of GG-I-526 and the following option exercises under that specification:

3.2 Material. Shall be Class 3, any type, or Class 4, any type except 440A, 440B or 440C, or a combination of the aforementioned.

3.3 Hardness. After proper stress relieving, Rockwell hardness shall be B-75 minimum and C-45 maximum for all component parts of the retractor.

3.4 Style, design and dimensions. Shall be a self-retaining abdominal retractor of the Balfour type, having a notched spreader bar to maintain the lateral blades in position, with six blades, including one large and one small supra-pubic blade, and two each large and small lateral blades. Shall be either Style 1 or Style 2. Dimensions, style, and design for style 1 shall be in accordance with Defense Personnel Support Center Drawing No. 1286; dimensions, style and design for style 2 shall be in accordance with Defense Personnel Support Center Drawing No. 20568.

3.5 Finish. Finish shall be mirror and satin, as shown on appropriate drawing.

3.6 Identification marking. In addition to GG-I-526 requirements, retractors of foreign manufacture shall also be marked stainless.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any tiem, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the stainless steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table II. The unit of product for examination shall be one retractor.

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Table II. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III. Unit of product for test purposes shall be one retractor.

Table III. Sampling for Tests

Component (unit of product)	Characteristic	Req.	Test procedure	Insp. level	AQL (percent defective)
				**	*
1-retractor	Composition analysis	3.2 & GG-I-526	GG-I-526		*
	Hardness	3.3 & GG-I-526	GG-I-526	S-2	*
	Dimensions	3.4	---	S-1	*
	Boil test	GG-I-526		S-1	*
	Copper sulfate	GG-I-525	GG-I-526	S-1	*

* Acceptance number shall be zero for all sample sizes.
** Sufficient amount from each lot of each metal used in the construction of the retractor shall be selected for composition analysis.

4.3 Examination. The retractor shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Examination, however, shall not be restricted to defects listed in Table IV. Any retractor in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for test sample, the lot represented by the sample shall be rejected.

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Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Instrument not constructed as specified.
102	Style and design not as specified.
103	Instrument not free of sharp edges, rough areas, burrs, cracks, grind marks or overlaps.
Minor	
201	Identification marking not present, correct, complete, or legible.
202	Finish not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements in accordance with GG-I-526 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One retractor, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Preservative shall not be applied.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method 1C-2.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, class domestic. Commercial colors are acceptable on intermediate cartons and closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

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5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The abdominal retractors shall be suitable for use as self-retaining retractors in certain types of lower abdominal surgery.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1 and 5.2)

6.3 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

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National Stock Number	Item Identification
6515-00-360-4910	RETRACTOR, ABDOMINAL, Balfour, Self-Retaining, 6 Blades

It does not include all types or styles as indicated by the title nor all types or styles which are commercially available.

6.4 Supersession data. This specification includes the requirements of Purchase Description No. 9, dated 12 December 1966 for FSN 6515-360-4910 RETRACTOR, ABDOMINAL, Balfour, Self-Retaining, 6 Blades.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1746

Review activities:

Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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